AKPOJOTOR OGHENECHOVWE EMMANUEL

Email: nou224068313@noun.edu.ng
LinkedIn: www.linkedin.com/in/databit365
GitHub: https://github.com/iMonie

Proposed Research Area:
AI-Driven Predictive Maintenance and Concept Drift Detection for Pipeline Integrity Monitoring

---

PROFESSIONAL PROFILE

Aspiring PhD researcher in Information Technology with strong focus on machine learning systems, adaptive intelligence, and AI-driven monitoring frameworks. My research interest centers on developing intelligent models capable of detecting evolving patterns in data streams and enabling predictive decision-making in real-world systems.

With over a decade of experience in ICT leadership within a healthcare environment, I bring practical expertise in system reliability, real-time monitoring, data management, and fault detection—providing a strong foundation for research in predictive maintenance and intelligent infrastructure systems.

RESEARCH INTERESTS

- Machine Learning Systems
- Adaptive AI & Concept Drift Detection
- Predictive Maintenance Systems
- Time-Series Data Analysis
- Intelligent Monitoring Systems
- Natural Language Processing
- Cloud-Based AI Systems

EDUCATION

Master of Science (M.Sc.) in Information Technology
National Open University of Nigeria — March 2026

Bachelor of Science (B.Sc.) in Information Technology
National Open University of Nigeria — March 2022

PROFESSIONAL EXPERIENCE

Chief Technical Officer (ICT Department)
Delta State University Teaching Hospital, Oghara, Nigeria
January 2010 – Present

Key Responsibilities:

- Manage and maintain enterprise-wide ICT infrastructure and hospital information systems
- Oversee network administration, cybersecurity, and data management operations
- Monitor mission-critical systems requiring high availability, reliability, and fault tolerance
- Support continuous system operations in real-time service environments
- Implement monitoring strategies for early fault detection and system optimization
- Lead deployment, upgrade, and maintenance of healthcare IT systems
- Supervise technical support services across multiple departments
- Coordinate infrastructure improvements and performance optimization initiatives

PUBLICATIONS

Akpojotor, O. E., & Obabunmi, R. (2026)
An End-to-End Machine Learning System for Real-Time Credit Card Fraud Detection with Interactive Deployment
Zenodo *(Open-access research repository)*
DOI: https://doi.org/10.5281/zenodo.20053984

Akpojotor, O. E. et al. (2026)
Multilingual Prompt Engineering as a Gateway to Agentic AI in Low-Resource Healthcare: Insights from Open Educational Manuals in Sub-Saharan Contexts
Zenodo *(Open-access research repository)*
DOI: https://doi.org/10.5281/zenodo.19533533

Akpojotor, O. E. et al. (2025)
Artificial Intelligence in Smart Cities: Optimizing Urban Planning and Resource Management
Zenodo *(Open-access research repository)*
DOI: https://doi.org/10.5281/zenodo.15208541

Akpojotor, O. E. et al. (2025)
Advancing Handwritten Digit Recognition: A Hybrid Deep Learning Framework with Attention Mechanisms
Zenodo *(Open-access research repository)*
DOI: https://doi.org/10.5281/zenodo.15397901

RESEARCH & TECHNICAL PROJECTS

Fraud Detection AI System

- Developed an end-to-end machine learning pipeline for fraud detection
- Addressed class imbalance using SMOTE
- Evaluated models including Logistic Regression, Random Forest, and XGBoost
- Built and deployed an interactive system for real-time prediction
  GitHub: https://github.com/iMonie/fraud-detection-ml

Exploratory Analysis of Distribution Shifts in Monitoring Data

- Simulated streaming data environments to model real-time monitoring systems
- Applied statistical methods (Kolmogorov–Smirnov test) for drift detection
- Implemented sliding window analysis for continuous evaluation
- Visualized evolving data patterns and system behavior
  GitHub: https://github.com/iMonie/shift-detection-study

MNIST Digit Classification (Deep Learning Project)

- Developed a Convolutional Neural Network (CNN) for handwritten digit recognition
- Demonstrated deep learning architecture design and evaluation techniques
- Linked to published research on hybrid deep learning frameworks
  GitHub: https://github.com/iMonie/mnistproject

AWS Cloud-Based Portfolio System

- Designed and deployed a serverless cloud-based application
- Integrated AWS S3, Lambda, API Gateway, and DynamoDB
- Implemented real-time visitor tracking functionality
  GitHub: https://github.com/iMonie/aws-cloud-portfolio

TECHNICAL SKILLS

Programming: Python, JavaScript, HTML/CSS
Machine Learning: Scikit-learn, TensorFlow, XGBoost
Data Analysis: Pandas, NumPy, Matplotlib
Cloud Computing: AWS (S3, Lambda, API Gateway, DynamoDB)
Tools: Git, GitHub, Gradio

PROFESSIONAL MEMBERSHIP

- Member, Computer Professionals Registration Council of Nigeria (CPN)